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Virtual signage using augmented reality or mixed reality

Abstract

An information system for displaying virtual signage in a medical facility includes a treatment device. A visual identifier is operably coupled to the treatment device. A controller is configured to communicate with a remote device having an image sensor for sensing the visual identifier within a field of detection. The controller is configured to recognize the visual identifier sensed by the remote device, determine device information associated with the visual identifier based on a configuration of the visual identifier, retrieve the device information relating to the treatment device associated with the visual identifier from an information source, and generate a virtual image including the device information configured to be viewed via the remote device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims priority to U.S. Provisional Patent Application No. 63/132,660, filed on Dec. 31, 2020, entitled “VIRTUAL SIGNAGE USING AUGMENTED REALITY OR MIXED REALITY,” the disclosure to which is hereby incorporated herein by reference in its entirety.

FIELD OF THE DISCLOSURE
(1) The present disclosure generally relates to virtual signage, and more particularly to virtual signage at a medical facility utilizing augmented reality or mixed reality.
SUMMARY OF THE DISCLOSURE
(2) According to one aspect of the present disclosure, an information system for displaying virtual signage in a medical facility includes a treatment device. A visual identifier is operably coupled to the treatment device. A controller is configured to communicate with a remote device having a sensor for sensing the visual identifier within a field of detection. The controller is configured to recognize the visual identifier sensed by the remote device, determine device information associated with the visual identifier based on a configuration of the visual identifier, retrieve the device information relating to the treatment device associated with the visual identifier from an information source, and generate a virtual image including the device information configured to be viewed via the remote device.
(3) According to another aspect of the present disclosure, an information system for a medical facility in which visual identifiers are associated with the medical facility where the information system includes a query member associated with each visual identifier. The query member is at least one of a patient identification feature, a treatment device, and a room environment. A controller is configured to communicate with a remote device and a server. The controller is configured to recognize the visual identifiers sensed by the remote device, associate the visual identifiers with information related to at least one of a patient, the treatment device, and the room environment based on the query member, retrieve the information from an information source, and generate a virtual image including the information to be communicated to a user via the remote device.
(4) According to yet another aspect of the present disclosure, a method of displaying information to a caregiver, including sensing a visual identifier positioned within a field of detection of a sensor of a remote device via at least one of an imager and an environmental sensor; recognizing the visual identifier; retrieving information relating to at least one of a room environment, a patient, and a treatment device based on the configuration of the visual identifier; generating a virtual image including the information; and displaying the virtual image via at least one of a display of the remote device and within a field of view of a user of the remote device.
(5) These and other features, advantages, and objects of the present disclosure will be further understood and appreciated by those skilled in the art by reference to the following specification, claims, and appended drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) In the drawings:

(2) FIG. **1** is a schematic diagram of a portion of a medical facility having an information system of the present disclosure;
(3) FIG. **2** is a block diagram of an information system for a medical facility, according to the present disclosure;
(4) FIG. **3** is a block diagram of an information system with a treatment device in wireless communication with a server, according to the present disclosure;
(5) FIG. **4** is a block diagram of an information system with a treatment device in wireless communication with a server, according to the present disclosure;
(6) FIG. **5** is representative of an authorized access interface of a remote device associated with an information system, according to the present disclosure;
(7) FIG. **6** is representative of a patient portal of a caregiver interaction system displayed on a remote device associated with an information system, according to the present disclosure;
(8) FIG. **7** is representative of a chat feature of a caregiver interaction system displayed on a remote device associated with an information system, according to the present disclosure;
(9) FIG. **8** is a schematic view of a caregiver using a remote device to recognize a visual identifier on a room plaque, according to the present disclosure;
(10) FIG. **9** is representative of an image overlaid on captured image data and displayed on a remote device, according to the present disclosure;
(11) FIG. **10** is a schematic view of a caregiver in a room environment and using a wearable remote device, according to the present disclosure;
(12) FIG. **11** is a schematic view of a virtual image including information associated with a medical bed projected for a caregiver to view with a wearable remote device, according to the present disclosure;
(13) FIG. **12** is a schematic view of a virtual image including patient information projected for a caregiver to view with a wearable remote device, according to the present disclosure;
(14) FIG. **13** is a schematic view of an updated image projected after interaction with the image of FIG. **12**; and
(15) FIG. **14** is a flow diagram of a method for displaying information to a caregiver, according to the present disclosure.

DETAILED DESCRIPTION

(16) The present illustrated embodiments reside primarily in combinations of method steps and apparatus components related to virtual signage using augmented reality or mixed reality. Accordingly, the apparatus components and method steps have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Further, like numerals in the description and drawings represent like elements.
(17) For purposes of description herein, the terms “upper,” “lower,” “right,” “left,” “rear,” “front,” “vertical,” “horizontal,” and derivatives thereof, shall relate to the disclosure as oriented in FIG. **1**. Unless stated otherwise, the term “front” shall refer to a surface closest to an intended viewer, and the term “rear” shall refer to a surface furthest from the intended viewer. However, it is to be understood that the disclosure may assume various alternative orientations, except where expressly specified to the contrary. It is also to be understood that the specific structures and processes illustrated in the attached drawings, and described in the following specification are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.
(18) The terms “including,” “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises

a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by "comprises a . . . " does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.

(19) Referring to FIGS. **1-14**, reference numeral **10** generally designates an information system for a medical facility **12** that includes a treatment device **14**. A data tag, such as a visual identifier **16**, is operably coupled to the treatment device **14**. The controller **18** is configured to communicate with a remote device **20** having a sensor **22** for sensing the visual identifier **16** within a field of detection **24**. The controller **18** is configured to recognize the visual identifier **16** sensed by the remote device **20**. The controller **18** is configured to determine device information **26** associated with the visual identifier **16** based on a configuration of the visual identifier **16** and retrieve the device information **26** relating to the treatment device **14** associated with the visual identifier **16** from an information source **28**. Additionally, the controller **18** is configured to generate a virtual image **30** that includes the device information **26**, which is viewable via the remote device **20**.

(20) The information system **10** is configured to retrieve information from one or more information sources **28** to present the information to a caregiver or another user. Each visual identifier **16** has a specific and unique configuration, which allows the controller **18** to recognize the visual identifier **16**. Based on the configuration of the visual identifier **16**, the controller **18** is configured to determine what information to retrieve and from where the information is to be retrieved. Further, each visual identifier **16** is associated with a specific query member **36**. Each query member **36** is an item, space, person, device, etc. that the caregiver is seeking information about. In various examples, the query members **36** are the treatment device **14**, a patient identification feature **38** worn by a patient, and a room environment **40** of the medical facility **12**. Each query member **36** has a specific data tag or visual identifier **16** with a specific configuration, which allows the information system **10** to determine the information to be obtained for the caregiver.

(21) Referring to FIG. **1**, the medical facility **12** includes multiple room environments **40**, such as patient rooms **40**A, surgical suites **40**B, imaging areas, waiting rooms, etc. on a number of floors **42**, illustrated as floors **42**A-**42**D. Each room environment **40** includes at least one treatment device **14** for treating or otherwise caring for a patient. In the configuration illustrated in FIG. **1**, the treatment devices **14** include a medical bed **44** within each patient room **40**A and a surgical table **46** in the surgical suite **40**B. The treatment devices **14** may communicate the device information **26**, as well as patient information **48**, to the information system **10**, as discussed further herein.

(22) Referring to FIG. **1**, as well as FIG. **2**, the information system **10** provides a caregiver with more convenient and efficient access to the device information **26** (FIG. **11**) and the patient information **48** (FIG. **9**), as well as other confidential or private information useful for caring for the patient. Traditionally, the device information **26** and the patient information **48** are included in hardware or handwritten notes and charts, which can be cumbersome, expensive, and have potential privacy concerns. The information system **10** for the medical facility **12** disclosed herein utilizes the remote device **20** to display virtual signage that may include the device information **26**, the patient information **48**, any additional information useful for the caregiver, or a combination thereof.

(23) The virtual signage (e.g., the virtual image **30**) is displayed using at least one of augmented reality and mixed reality. The use of augmented reality or mixed reality provides a convenient and efficient method for accessing information stored in a variety of locations or systems while maintaining the confidentiality of the information. Additionally, the use of augmented reality or mixed reality restricts access to the information to those caregivers and other medical professionals with authorized access to the information.

(24) The remote device **20** may have a variety of configurations that may determine whether augmented reality or mixed reality is utilized to view the information. Augmented reality overlays virtual objects on a real-world environment to enhance the real-world environment. Generally, the

real-world environment includes captured image data and the virtual objects are overlaid on the captured data so a user can view both the real-world environment and the virtual objects together on a display or device. The user interacts with the real-world environment while digital or virtual content is added.

(25) In comparison, mixed reality goes beyond augmented reality and allows the user to interact with the virtual objects. The virtual objects are overlaid on the real-world environment; however, the virtual objects respond and react to the user as a real object would as set forth in further detail herein. Generally, the virtual objects are projected into the real-world environment that the user views using a wearable device or display. The information system **10** may use one or both of augmented reality and mixed reality to display information to the caregiver, as discussed further herein.

(26) Referring still to FIG. **2**, the controller **18** of the information system **10** may be included at least partially in a local server **50** of the medical facility **12**, a remote server **52**, or a combination thereof. The controller **18** includes a processor **54**, a memory **56**, and other control circuitry. Instructions or routines **58** are stored within the memory **56** and executable by the processor **54**. The control circuitry may also include communication circuitry **60** to permit communication via a communication network **62** or various protocols for wireless communication in combination with a wired network, as discussed further herein. Additionally, the routines **58** may include operating instructions to enable the various processes and methods described herein.

(27) The local server **50** is in communication with at least one of the remote device **20**, the treatment device **14**, and the remote server **52** via the communication network **62**. The communication network **62** may be part of an overall facility network of the medical facility **12**. The facility network may include a combination of wired connections (e.g., Ethernet **66**, as illustrated in FIG. **3**) as well as wireless networks, which may include the wireless communication network **62**. The communication network **62** may include a variety of electronic devices, which are configured to communicate over various wired or wireless communication protocols.

(28) In the illustrated configuration of FIG. **2**, the local server **50** is in wireless communication with the treatment devices **14**, the remote device **20**, and the remote server **52**. The communication network **62** may include a wireless router through which the remotely accessed treatment devices **14** and the remote device **20** may be in communication with one another, as well as the local server **50** or the remote server **52** via the network.

(29) The communication network **62** may be implemented via one or more direct or indirect, non-hierarchical communication protocols, including, but not limited to, Bluetooth®, Bluetooth® low energy (BLE), Thread, Ultra-Wideband, Z-Wave, ZigBee®, etc. Additionally, the communication network **62** may correspond to a centralized or hierarchal communication network **62** where one or more of the treatment devices **14** or the remote device **20** communicate via the wireless router (e.g., a communication routing controller). Accordingly, the communication network **62** may be implemented in a variety of communication protocols in various combinations, including, but not limited to global system for mobile communication (GSM), general packet radio services, code division multiple access, enhanced state GSM environment, fourth generation (4G) wireless, fifth generation (5G) wireless, Wi-Fi, world interoperability for microwave access (WiMAX), local area network (LAN), Ethernet **66**, etc. By flexibly implementing the communication network **62**, the various treatment devices **14** and the remote device **20** may be in communication with one another and the remote server **52** directly via the wireless communication network **62** or via a cellular data connection.

(30) Referring still to FIG. **2**, the remote device **20** may be configured as a handheld device **80**. The handheld device **80** may be a phone, a laptop, a tablet, or other portable handheld devices **80** associated with the medical facility **12** or the caregiver. In certain aspects, the handheld device **80** may be a personal caregiver device, such as a personal phone, which is convenient or efficient for the caregiver as the caregiver conducts rounds between the various patient rooms **40**A (FIG. **1**).

(31) The handheld device **80** includes a control unit **82** having a processor **84**, a memory **86**, and other control circuitry. Instructions or routines **88** are stored within the memory **86** and executable by the processor **84**. The handheld device **80** may include communication circuitry **90** for communicating with at least one of the local server **50**, the remote server **52**, and the treatment devices **14** through the communication network **62**. The control circuitry may also include image processing circuitry for processing image data captured by the handheld device **80**.
(32) The handheld device **80** generally includes the sensor **22** configured as an imager **92**, also referred to as an image sensor, defining the field of detection **24** (FIG. **8**). The imager **92** is generally a forward facing imager or camera on a backside of the handheld device **80** relative to a display **96**. The handheld device **80** may also include an additional image sensor or imager **98**, which may be rearward (e.g., on an opposing side relative to the display **96**) or forward facing (e.g., on a same side as the display **96**) depending on the configuration of the handheld device **80**. The imagers **92**, **98** may each be any practicable type of image-based sensor, such as a charge coupled device, a metal oxide semiconductor imager, or any type of color or black-and-white camera. The imager **92** captures data from the field of detection **24**, while the imager **98** captures image data within a separate field of detection. The data captured by the imagers **92**, **98** generally includes image data, such as at least one of a picture, video, real-time streaming of data, other transmissions of image data, or combinations thereof. The image data may be a single image or multiple images.
(33) The imagers **92**, **98** may be adjustable, which may also adjust the field of detection **24** to be broader, narrower, positionally shifted, or any combination thereof. The imagers **92**, **98** may receive a signal from the controller **18** based on the data or a user input to adjust an aspect of the imagers **92**, **98**. For example, the imager **92** may be adjusted to change the scope of the field of detection **24**. It is contemplated that each imager **92**, **98** includes one or more lenses, which may be adjusted to change the sharpness or quality of the data obtained by the imagers **92**, **98**, respectively. Generally, the imager **92** captures image data relating to the visual identifier **16** (FIG. **8**), which is then communicated to the controller **18** to retrieve the information associated with the visual identifier **16**. The associated information is then conveyed to the caregiver via the display **96** of the handheld device **80** using augmented reality. The additional imager **98** may be utilized for facial recognition of the caregiver for providing access to the information as described further herein.
(34) Additionally or alternatively, the remote device **20** may be configured as a wearable device **100**. Generally, the wearable device **100** is configured as glasses or another head-mounted display. The wearable device **100** includes a control unit **102** that has a processor **104**, a memory **106**, and other control circuitry. Instructions or routines **108** are stored within the memory **106** and executable by the processor **104**. The wearable device **100** includes communication circuitry **110** for communicating via the communication network **62**. The control circuitry may also be configured to process sensed information obtained by the wearable device **100**.
(35) The wearable device **100** may be utilized to display information using augmented reality and/or mixed reality. The wearable device **100** may display information via augmented reality in a similar manner as discussed in relation to the handheld device **80**. To utilize mixed reality, the wearable device **100** includes the sensor **22** configured as environmental sensors **112** for sensing a variety of environmental information in the surrounding environment of the caregiver. The environmental sensors **112** each define the field of detection **24**, respectively, that extends from the wearable device **100** and away from the caregiver. The environmental sensors **112** sense the presence of objects within the surrounding environment, the position and distance to the objects, the depth of the object, lighting information, a combination thereof, etc.
(36) For example, the environmental sensors **112** may include infrared cameras or Light Detection and Ranging (LIDAR) emitters and detectors to capture depth or range in the surrounding environment. The environmental sensors **112** may also include multiple sensors, such as an infrared sensor or Red, Green, Blue (RGB) cameras that sense information about the movement of the user, such as the position, orientation, and motion of the user within the environment. Further, the

environmental sensors **112** may sense the interaction of the caregiver with the sensed objects and/or with the virtual image **30**.

(37) The wearable device **100** also includes the sensor **22** configured as user sensors **114**, which are generally configured to monitor or sense additional information about the user (e.g., the caregiver). The user sensors **114** may include an inertial-movement unit that monitors the movement of the head of the caregiver. Additionally or alternatively, the user sensors **114** may include eye-tracking sensors to track the position and movement of eyes of the caregiver. The eye-tracking sensors may track a focus of the caregiver (e.g., a focus direction) and define a field of view **116** (FIG. **10**) of the caregiver corresponding with a line of sight into the surrounding environment.

(38) The direction of the focus of the caregiver may be determined by measuring an eye-ellipse of the caregiver. The eye-ellipse is a graphical device that represents the approximation of the eye location distribution of the caregiver as a multidimensional normal density distribution. The eye-ellipse results in a set of lines that isolate an ellipse area, which may account for about 90% of eye positions. The direction of focus of the caregiver may be utilized by the information system **10** to display or project the virtual image **30** using mixed reality through a projector **118** of the wearable device **100**. Further, the user sensors **114** may include at least one gesture sensor to track position, movement, and gestures of the caregiver, which may determine the interaction of the caregiver with the virtual image **30**.

(39) Referring still to FIG. **2**, the remote device **20** conveys information to the caregiver. The controller **18** retrieves the information from various information sources **28** including multiple locations and systems. The information sources **28** may include the local server **50**, the remote server **52**, an electronic medical record **120**, a caregiver interaction system **122**, a nurse call system **124**, other systems of the medical facility **12**, or a combination thereof. For example, the controller **18** is in communication with remote server **52** to obtain the patient information **48** from the electronic medical record **120** and/or to obtain facility protocols **126**. Additionally or alternatively, the electronic medical records **120**, the facility protocols **126**, or a combination thereof may be stored on the local server **50**.

(40) The electronic medical record **120** may be associated with the patient and include current and historical information relating to demographics, allergies, infections, treatments, medications, medical history, etc. of the patient. The caregiver interaction system **122** may include information communicated between caregivers, recent updates, shared data, etc. that may or may not be ultimately stored in the electronic medical record **120**. Additionally, the nurse call system **124** may include communication between the patient and the caregiver including, number of calls, type of calls, substance of calls, etc. The information and software for the various systems may be stored on the local server **50**, the remote server **52**, the handheld device **80** (such as the personal caregiver device), or a combination thereof. The controller **18** communicates with the handheld device **80**, the remote server **52**, and/or the local server **50** to retrieve information associated with the patient to subsequently convey to the caregiver via the remote device **20**.

(41) Additionally or alternatively, in the illustrated example of FIG. **2**, the remote server **52** stores the facility protocols **126**, which may relate to, for example, safety features, treatment processes, device protocols, etc. The facility protocols **126** may determine or contribute to the information communicated to the caregiver. For example, if a certain safety protocol is not initiated but should be initiated based on the patient information **48**, the information conveyed to the caregiver may prompt activation of the safety protocol.

(42) The information system **10** also retrieves the device information **26** and/or the patient information **48** from the treatment device **14**. Generally, a variety of treatment devices **14** may be utilized for treating and caring for the patient while the patient is at the medical facility **12**. Each treatment device **14** includes a control unit **130** that has a processor **132**, a memory **134**, and other control circuitry. Instructions or routines **136** are stored within the memory **134** and executable by the processor **132**. The treatment device **14** generally includes a sensor assembly **140** configured to

sense information about the patient to be communicated to the information system **10**.

(43) Each treatment device **14** may include communication circuitry **138** for communicating via the communication network **62**. The information system **10** may communicate directly with the treatment device **14** without communicating through the communication network **62**. Additionally or alternatively, the treatment device **14** may communicate with the local server **50**, and the information system **10** may obtain the information from the local server **50**.

(44) Referring to FIGS. **3** and **4**, exemplary configurations of the communication between the treatment device **14** and the local server **50** are illustrated. For example, as illustrated in FIG. **3**, the treatment device **14** may be configured to communicate with a wireless access point transceiver **150**, which is coupled to Ethernet **66** of the medical facility **12**. The communication network **62** provides for bidirectional communication between the treatment device **14** and the wireless access point transceiver **150**. The treatment device **14** is associated with a network interface unit **152**. Multiple network interface units **152** may be provided in various locations of the medical facility **12**.

(45) Each treatment device **14** and each network interface unit **152** is assigned a unique identification (ID) code, such as a serial number. Various components of the information system **10** (e.g. the local server **50**, the remote device **20**, etc.) may include software (e.g. routines **58**, **108**, **136**) that operate to associate the ID of the treatment device **14** with the network interface unit ID data to locate each treatment device **14** within the medical facility **12**. Each network interface unit **152** includes a port **154** for selectively coupling with Ethernet **66**. When the network interface unit **152** is coupled with Ethernet **66**, the network interface unit **152** communicates ID data to treatment device **14**, which then wirelessly communicates ID data for the treatment device **14** and the network interface unit **152** to the wireless access point transceiver **150**. The wireless access point transceiver **150** communicates bidirectionally with Ethernet **66** via a data link **156**. The local server **50** is in communication with Ethernet **66** to receive the data or information from the treatment device **14**.

(46) Additionally or alternatively, as illustrated in FIG. **4**, the treatment device **14** may be capable of communicating wirelessly via a wireless communication module **158**. The wireless communication module **158** generally communicates via an SPI link with circuitry of the associated treatment device **14** (e.g., the communication circuitry **138**) via a wireless 802.11b link and the wireless access point transceiver **150**. Multiple wireless access point transceivers **150** may be also located throughout the medical facility **12**. The wireless access point transceivers **150** are generally coupled to Ethernet switches **160** via 802.3 links. It is contemplated that the wireless communication modules **158** may communicate with the wireless access point transceivers **150** via any of the wireless protocols disclosed herein. Additionally or alternatively, the Ethernet switches **160** generally communicate with Ethernet **66** via an 802.3 link, and Ethernet **66** is also in communication with the local server **50**, allowing information and data to be communicated between the local server **50** and the treatment device **14**.

(47) Referring again to FIG. **2**, as well as FIGS. **5-7**, the controller **18** may be associated with the caregiver interaction system **122** accessible via an application interface on, for example, the remote device **20**, a facility device (such as at a nurse call station), or other devices. The caregiver interaction system **122** may be accessible via an application or software on the remote device **20**. The caregiver interaction system **122** provides communication between some or all of the caregivers associated with the patient during the treatment or stay of the patient at the medical facility **12**. The caregiver interaction system **122** may also provide a process for communicating information about the patient between caregivers, as well as to the electronic medical record **120** (FIG. **2**).

(48) In the illustrated example of FIGS. **5-7**, the handheld device **80** is illustrated with information conveyed to the caregiver via the display **96**. As illustrated in FIG. **5**, an authorized access interface **168** is provided on the handheld device **80** to receive the credentials or other identification

information of the caregiver. The caregiver may input access information (e.g., user name and password) to gain access to the caregiver interaction system **122** and additional facility systems, such as the information system **10**.

(49) Additionally or alternatively, the handheld device **80** may have a touch identification feature **170** for recognizing a fingerprint (e.g., identification information) of the caregiver. The touch identification feature **170** includes a sensor or imager for sensing the fingerprint of the caregiver. The sensed fingerprint may be compared to a stored image within the memory **86** of the handheld device **80** to confirm access to the caregiver interaction system **122** and the information system **10**. The information system **10** may also store images of caregiver fingerprints and the handheld device **80** may communicate with the information system **10** to confirm access of the caregiver. At least one of the controller **18** of the information system **10** and the handheld device **80** includes routines **58**, **88** for comparing the sensed fingerprint with stored data to confirm authorization and access of the caregiver.

(50) In additional examples, the handheld device **80** may include the additional imager **98**, which may be utilized to obtain identification information, such as facial recognition or eye recognition (e.g., iris authentication), to grant access to the caregiver. The additional imager **98** may be a rearward facing imager on the same side of the handheld device **80** as the display **96**. The imager **98** may be configured to capture image data of the face of the caregiver for identification and authorization purposes. The captured image data of the face of the caregiver may be compared to stored images in the handheld device **80** or in the controller **18** of the information system **10**. At least one of the controller **18** of the information system **10** and the handheld device **80** includes routines **58**, **88** for comparing the detected image of the caregiver with the stored data to determine access of the caregiver.

(51) As best illustrated in FIG. **6**, the caregiver is able to access the caregiver interaction system **122** via the authorized access interface **168**, the touch identification feature **170**, the facial recognition, or eye recognition. Once access is granted to the caregiver interaction system **122**, the caregiver has access to a portal **174** that displays a variety of information about the patient or patients associated with the caregiver. The portal **174** may display messages from other caregivers, alerts, patient information **48**, facility protocols **126**, and other information helpful for providing care for the patient. The portal **174** may provide updates to the caregiver about the patient organized in a single location.

(52) The information and systems accessible by the caregiver may depend on the level of access of the caregiver or the role of the caregiver. For example, the information system **10**, may enable features depending on the role of the individual or user to provide role based access controls (RBAC) for the individual that is signed into the information system **10**. In such examples, a nurse may view certain patient details or have different options that a technician might not see.

(53) As best illustrated in FIG. **7**, the caregiver interaction system **122** provides a chat feature **176** for caregivers to communicate with one another about the patient (e.g., share the patient information **48**) and the treatment for the patient. The information shared in the chat feature **176** may not be information that is stored in the electronic medical record **120** (FIG. **2**). Accordingly, the caregiver interaction system **122** is another system within the medical facility **12** that includes the patient information **48** and/or the device information **26** that may be retrieved by the controller **18** and conveyed to the caregiver as part of the information system **10**. Moreover, the virtual image **30** generated by the controller **18** of the information system **10** may be communicated to other caregivers associated with the patient via the chat feature **176**.

(54) It is contemplated that the caregiver interaction system **122** may also be viewed or accessed via the wearable device **100**. In such configurations, the controller **18** may generate and project the virtual image **30** having the authorized access interface **168**. The sensors **112**, **114** may sense identification information based on the movement or gestures of the caregiver. The sensors **112**, **114** may also sense or scan an identification badge, which provides the identification information

utilized for accessing the various systems. Additionally or alternatively, the user sensors **114** may be configured for facial recognition and/or iris authentication to grant access to the information system **10**.
(55) Referring again to FIG. **2**, as well as FIGS. **8** and **9**, the information system **10** is illustrated using augmented reality to display information through the display **96** of the handheld device **80**. The caregiver arranges the handheld device **80** to position the imager **92** relative to the visual identifier **16**. The imager **92** captures image data (e.g., real-world data) in the field of detection **24** and communicates the captured image data to the control unit **82**. The image data includes the visual identifier **16**, which is illustrated on a room plaque **180** (e.g., the query member **36**) outside the patient room **40**A in the example in FIG. **8**. The visual identifier **16** may be a barcode, a quick response (QR) code, a pattern, an image, etc. identifiable and distinguishable by the information system **10**.
(56) The image data relating to the visual identifier **16** is communicated from the handheld device **80** to the controller **18** of the information system **10**. The controller **18** includes at least one routine **58** for analyzing the visual identifier **16** and determining the information associated with the visual identifier **16** to be retrieved. The controller **18** may process the image data to determine the specific configuration of the visual identifier **16**, which allows the controller **18** to determine at least one of the query member **36** with which the visual identifier **16** is associated and the information to be retrieved. Further, the controller **18** retrieves the information from at least one information source **28**, including at least one of the electronic medical record **120**, the facility protocols **126**, the treatment device **14**, the local server **50**, the remote server **52**, and the caregiver interaction system **122** via the wired or wireless protocols disclosed herein.
(57) The controller **18** retrieves and compiles the data or information associated with the visual identifier **16** and generates the virtual image **30**, which includes some or all of the retrieved information. Some information may be restricted based on the access level of the caregiver. For example, the information system **10** may be the RBAC system, which provides different information to different roles at the medical facility **12**. Certain caregivers may utilize more or different information to treat or care for the patient. The information presented to the caregiver in the virtual image **30** may be roles-based, allowing for select information to be presented to the caregiver based on the specific role or position of the caregiver. The virtual image **30** may include text, graphics, images, charts, graphs, etc. that convey information to the caregiver. The image data may be communicated to the controller **18** and the controller **18** overlays the virtual image **30** on the image data and communicates the combined image data to the handheld device **80** to be viewed on the display **96**. Alternatively, the virtual image **30** may be communicated to the handheld device **80** and the handheld device **80** may combine the virtual image **30** with the image data.
(58) For example, in the illustrated configuration of FIGS. **8** and **9**, the caregiver positions the handheld device **80** to capture the visual identifier **16** on the room plaque **180** within the field of detection **24**. The imager **92** captures the image data, including the visual identifier **16**, the room plaque **180**, and a room number **182** within the field of detection **24**. The controller **18** analyzes the visual identifier **16** within the image data and associates the visual identifier **16** with the patient information **48**, the device information **26**, or a combination thereof based on the specific visual identifier **16**. The controller **18** retrieves the patient information **48** and the device information **26** and generates the corresponding virtual image **30**.
(59) Referring still to FIGS. **8** and **9**, the virtual image **30**, overlaid on the captured image data of the room plaque **180**, is displayed via the display **96** of the handheld device **80**, as best illustrated in FIG. **9**. Accordingly, the virtual image **30** augments the real-world view of the room plaque **180** by displaying the virtual patient information **48** over the real-world image data. The patient information **48** includes information that may be relevant or important for the caregiver prior to entering the patient room **40**A. The patient information **48** may include, for example, a patient name, allergies, current or historical medications, medication dose information, risk information,

alerts associated with the patient, as well as any other information that may be useful for the caregiver. The illustrated patient information **48** is exemplary and not meant to be limiting. The virtual image **30** may replace or supplement a traditional patient chart. Further, the location of the visual identifier **16** on the room plaque **180** is also exemplary.

(60) It is contemplated that each room environment **40** within the medical facility **12** includes the room plaque **180**, which may include the respective visual identifier **16**. Additionally or alternatively, a surface, such as on a wall, within the room environment **40** may include the visual identifier **16**. The visual identifier **16** associated with the room environment **40** may be utilized for conveying information about the room environment **40** (e.g., lighting conditions, temperature, etc.), procedures, treatments, the patient, schedule, or a combination thereof. Accordingly, the room environment **40** may also be an example of the query member **36** in which the caregiver seeks information. Further, it is also contemplated that the alerts or alarms presented in the virtual Image **30** may be associated with the nurse call system **124**. The nurse call system **124** indicates to the caregiver that the patient is in need of assistance.

(61) The visual identifier **16** may be changed or adjusted for each patient. In such examples, the visual identifier **16** may be replaced when the patient is no longer in the specific room environment **40**. Alternatively, the information associated with the visual identifier **16** may be changed or adjusted by changing the routines **58** of the information system **10**. In such examples, the visual identifier **16** remains at the select room environment **40**, but the controller is re-programmed (e.g., via new or adjusted routines **58**) to associate new or changed information with the visual identifier **16**.

(62) Referring again to FIG. **2**, as well as FIGS. **10-13**, the patient room **40**A is illustrated with a variety of treatment devices **14** including the medical bed **44**, a mattress **184**, an oxygen therapy device **186**, and a vital signs monitor **188**, which may each be an exemplary configuration of the query member **36**. The illustrated treatment devices **14** are exemplary and not meant to be limiting. Other treatment devices **14**, such as a microclimate management system associated with the mattress **184**, a pneumatic system associated with the mattress **184**, sequential compression devices, sub-epidermal moisture (SEM) scanners, and/or the surgical table **46** may be included in the patient room **40**A, the surgical suite **40**B, or elsewhere in the medical facility **12** without departing from the teachings herein.

(63) Each treatment device **14** includes the visual identifier **16**, which is associated with the specific treatment device **14**. The visual identifiers **16** in each location (e.g., each treatment device **14**, each room plaque **180** within the medical facility **12**, etc.) are different to allow the controller **18** to associate specific information with each visual identifier **16**. The controller **18** is configured to analyze the location information based on the configuration of the visual identifier **16** to determine the information associated with each specific visual identifier **16**.

(64) It is contemplated that other objects within the room environment **40** may also include data tags **16** or visual identifiers **16**. For example, when at the medical facility **12**, the patient generally wears the patient identification feature **38** (e.g., an identification bracelet **38**). The identification bracelet **38** may include the visual identifier **16**. In another example, a wall within the room environment **40** may have a visual identifier **16** that conveys information about the room environment **40**, such as temperature or lighting information. Each of these objects that includes the specific visual identifier **16** is an exemplary configuration of the query member **36**, which the caregiver is seeking information about via the information system **10**.

(65) The caregiver may use the handheld device **80** to view the information associated with the visual identifiers **16**, as previously discussed herein, and/or may use the wearable device **100** to view the information using augmented reality or mixed reality. When using the wearable device **100** for mixed reality, the caregiver uses the environmental sensors **112** to obtain image data or other data relating to the surrounding environment. The environmental sensors **112** sense or the visual identifier **16** within the field of view **116** and communicate the data relating to the visual

identifier to the control unit **102**. The data may be communicated as image data or other data depending on the type of sensor.

(66) The control unit **102** communicates the information relating to the visual identifier **16** to the controller **18** of the information system **10**, which analyzes the visual identifier **16** and retrieves the associated information (e.g., the device information **26**, the patient information **48**, etc.). The controller **18** generates the virtual image **30** with the associated information and communicates the virtual image **30** to the wearable device **100**. The virtual image **30** is then projected via the projector **118** into the field of view **116** of the caregiver. In this way, the caregiver may view his or her surrounding environment with the virtual image **30** incorporated into the surrounding environment.

(67) The device information **26** may include information from the treatment device **14** that relates to the operation of the treatment device **14** and/or information about the patient using the treatment device **14**. In the example illustrated in FIG. **11**, the treatment device **14** is configured as the medical bed **44**. The medical bed **44** includes the sensor assembly **140**, which may include sensors that sense information about the status of the medical bed **44** and sensors that sense information about the status of the patient. The information sensed by the sensor assembly **140** is generally conveyed to the caregiver via the information system **10** in the virtual image **30**. The device information **26** for the medical bed **44** may include, for example, a type of medical bed **44**, functions of the medical bed **44**, safety states in accordance with the facility protocols **126**, ongoing alerts, active therapies, current positions, activated protocols, etc.

(68) In various aspects, the sensor assembly **140** may sense a variety of information about the status of the medical bed **44**. For example, the sensor assembly **140** may sense whether a braking system of the medical bed **44** is properly initiated. In another example, the medical bed **44** includes an obstacle detection system, and the sensor assembly **140** senses objects within a movement path of a lift system that raises and lowers a support surface of the medical bed **44**. Additionally or alternatively, the medical bed **44** generally articulates between different positions (e.g., head elevated, foot elevated, etc.) and the sensor assembly **140** may sense the position of the medical bed **44**. In an additional example, the medical bed **44** or components coupled to the medical bed **44** may be powered by a battery. The sensor assembly **140** may sense additional components coupled with the medical bed **44**, as well as a charge level of the medical bed **44** or the associated components.

(69) The sensor assembly **140** may also sense information about the patient on the medical bed **44**. For example, the medical bed **44** may include a monitoring system, which monitors a position and/or movement of the patient on the support surface. The monitoring system may include pressure sensors (e.g., of the sensor assembly **140**) that monitor the weight distribution of the patient related to predetermined movement thresholds and/or relative to a predetermined center of gravity. The control unit **130** may compare subsequent information from the pressure sensors to determine the movement of the patient. Additionally, the patient information **48** from the medical bed **44** may include health metrics of the patient, such as heart rate or respiration rate, sensed via generally contact-free patient monitoring.

(70) The information from the treatment device **14**, including information from the sensor assembly **140**, may be included in the virtual image **30** generated by the controller **18**. Further, this information may be compared to the facility protocols **126**. In such configurations, the controller **18** retrieves or processes the facility protocols **126** from the remote server **52** and compares the information from the treatment device **14** and the patient information **48** with the facility protocols **126**. The controller **18** may then generate the virtual image **30** with information about the facility protocols **126**, such as an alert that the facility protocols **126** are not currently being followed and a prompt to adjust the treatment device **14** to follow the facility protocol **126**.

(71) Referring still to FIGS. **2** and **10-12**, there are multiple facility protocols **126** that may be utilized by the information system **10**. For example, the facility protocols **126** may include a fall

risk protocol. When the fall risk protocol is initiated, the patient may not exit the medical bed **44** or other support apparatus or surface without the assistance of the caregiver due to a heightened risk that the patient may fall or be injured. The monitoring system of the medical bed **44** monitors the movement of the patient and if the patient is sensed proximate to an edge of the medical bed **44**, the virtual image **30** may include an alert or alarm to notify the caregiver of the risk to the patient, as illustrated in FIG. **11**.

(72) Additionally or alternatively, if the monitoring systems senses movement of the patient and a siderail of the medical bed **44** is lowered (as sensed by the sensor assembly **140**), the virtual image **30** may include an alert that the patient may be trying to exit the medical bed **44**. When using mixed reality, the virtual image **30** may include selectable features **194**. The caregiver may interact with the selectable features **194** within the virtual image **30**. The selectable features **194** may activate certain protocols or adjust certain aspects of the medical bed **44**. For example, as illustrated in FIG. **11**, the caregiver may select whether to raise the side rails of the medical bed **44**. The selection or interaction with the virtual image **30** may be sensed by at least one of the environmental sensors **112** and the user sensors **114** and compared to the virtual image **30**. The wearable device **100** may then communicate the command or selection of the user to the medical bed **44**.

(73) As illustrated in FIG. **11**, the virtual image **30** is showing information related to alarms. The virtual image **30** may be adjusted or changed in response to the movement of the caregiver. For example, the caregiver may swipe his hand in a left-to-right gesture, which may adjust the virtual image **30** to the “Bed Status” information. Additionally or alternatively, the headings or titles in the virtual image **30** (e.g., “Bed Date,” “Risk Protocols,” “Bed Status,” “Alarms,” etc.) may be selectable features **194**. The caregiver may point to the heading to change the virtual image **30** to include or display new or additional information. The environmental sensors **112**, the user sensors **114**, or both may track the movement and/or focus of the caregiver to sense the gesture or selection of the caregiver. The movement and/or the focus of the caregiver may be analyzed by the control unit **102** and/or the controller **18** and compared to the virtual image **30** to determine the selection or the input from the caregiver.

(74) In another example, the facility protocols **126** may include a pulmonary risk protocol. When the pulmonary risk protocol is activated, the sensor assembly **140** may monitor the head position of the patient. Generally, the head position is elevated at least about 30° relative to a flat position. If the head of the patient is not elevated, the virtual image **30** may include an alert that the head of the patient should be elevated and may also include a prompt to adjust the medical bed **44** to elevate the head of the patient, which may be selected via the virtual image **30**. An adjustment assembly of the medical bed **44** may be activated in response to the selection in the virtual image **30** to adjust the elevation angle of the head end of the medical bed **44**.

(75) Referring again to FIGS. **2**, **12**, and **13**, the treatment device **14** may be the vital signs monitor **188**, which generally communicates the patient information **48** to the information system **10**. It is contemplated that the vital signs monitor **188** may also communicate device information **26**, such as a status of the vital signs monitor **188**, alerts regarding the function of the vital signs monitor **188**, etc. The vital signs monitor **188** may include the sensor assembly **140** having a variety of physiological sensors to obtain vital signs information from the patient, including, but not limited to, core body temperature or skin temperature, pulse rate, heart rate, blood pressure, respiratory rate, body weight, other body signs such as end-tidal CO.sub.2, SpO.sub.2 (saturation of oxygen in arterial blood flow), and other indicators of the physiological state of the patient. The caregiver may utilize the wearable device **100** to view the information associated with the visual identifier **16** coupled to the vital signs monitor **188**.

(76) As illustrated in FIG. **12**, the virtual image **30** includes the patient information **48** relating to blood pressure, heart rate, and body temperature. The illustrated virtual image **30** includes an alert feature **196** proximate the body temperature information. The caregiver may interact with the

virtual image **30** to select the alert feature **196** (e.g., an exemplary selectable feature **194**), which changes the virtual image **30** to display the alert information illustrated in FIG. **13**.

(77) When using the wearable device **100**, the caregiver may interact with and manipulate the virtual image **30**. The environmental sensors **112** and/or the user sensors **114** may sense the movement or gestures of the caregiver. The control unit **102** of the wearable device **100** may compare the movement of the caregiver with the virtual image **30** projected from the projector **118**. The control unit **102** may identify the selection of the caregiver and communicate the selection to the controller **18** so the controller **18** may generate the subsequent or updated virtual image **30** in response to the selection. The updated virtual image **30** may then be communicated to the control unit **102** and projected into the field of view **116** of the caregiver.

(78) Referring again to FIGS. **10-13**, as the caregiver moves about the patient room **40**A, or any other room environment **40**, the caregiver may see multiple virtual images **30** depending on the field of view **116**. The wearable device **100** monitors the head and eye position of the caregiver to track the focus of the caregiver. The virtual images **30** projected into the field of view **116** of the caregiver may be adjusted relative to the caregiver as a real-world object would. For example, as the caregiver moves closer to the virtual image **30**, the virtual image **30** appears larger as if the caregiver is approaching a real object. The caregiver may move, adjust, and otherwise manipulate the virtual image **30** when using the wearable device **100**. It is contemplated that each visual identifier **16** within the patient room **40**A may also be sensed with the handheld device **80**, which would allow the caregiver to view the device information **26** and the patient information **48** via the display **96** using augmented reality.

(79) Referring to FIGS. **1-13**, the caregiver may use augmented reality, mixed reality, or both to view a variety of information relating to the patient, the treatment device **14**, the room environment **40**, etc. Visual identifiers **16** or other data tags **16** may be disposed in a variety of locations throughout the medical facility **12** on a variety of query members **36**. Each visual identifier **16** is unique and is associated with specific information, allowing the information system **10** to retrieve the information associated with certain visual identifiers **16** for the caregiver. The visual identifiers **16** associated with the patient may be updated via the information system **10** when the patient is discharged or may be replaced. The remote device **20** is in communication with the information system **10** to display the virtual image **30** generated by the controller **18** to the caregiver. The information system **10** provides an efficient and convenient process for the caregiver to view information that is generally stored in multiple locations.

(80) In augmented reality examples, the virtual image **30** that includes the added information (e.g., the device information **26**, the patient information **48**, etc.) is a generated image overlaid on captured or sensed image data and displayed together to the caregiver on the display **96**. The virtual image **30** augments the real-world image obtained by the remote device **20**. In mixed reality examples, the virtual image **30** is a generated image projected from the remote device **20** into the field of view **116** of the caregiver. The virtual image **30** is incorporated into the real-world surroundings of the caregiver, and the caregiver may manipulate or interact with the virtual image **30**. The caregiver, who has been granted access to the information system **10**, may view the information in the virtual image **30**, but others around the caregiver or others in the medical facility **12** may not view the information without obtaining access through the remote device **20**. In this way, the information system **10** maintains heightened privacy of the information in the virtual image **30**.

(81) Additionally or alternatively, different caregivers may have different levels of access. When the caregiver is identified by the information system **10**, different information may be included in the virtual image **30** to include the information for which the caregiver has access. Additional authorized information may not be included in the virtual image **30**. It is contemplated that a notification may be presented to the caregiver that access is unauthorized or limited.

(82) Referring to FIG. **14**, as well as FIGS. **1-13**, a method **200** of displaying information to the

caregiver starts at **202** with activation of the remote device **20** and proceeds to step **204** where the remote device **20** is positioned proximate to the room plaque **180** or other location outside of a room environment **40**. The remote device **20** is oriented to sense the first or initial visual identifier **16**. The visual identifier **16** may be any symbol, barcode, QR code, text, graphic, characters, codes, or data that may be optically recognized. In examples using the handheld device **80**, the handheld device **80** is positioned such that the field of detection **24** of the imager **92** includes the visual identifier **16**. In wearable device **100** examples, the caregiver positions his or her head such that the visual identifier **16** is within the field of detection **24** of the environmental sensors **112**, which may generally align with the field of view **116** of the caregiver. Generally, if the caregiver is positioned directly in front of the room plaque **180**, the environmental sensors **112** can sense the visual identifier **16**.

(83) Further, in step **204**, the remote device **20** may sense the surrounding environment and/or capture image data. Additionally, in step **204**, the remote device **20** senses the visual identifier **16** and communicates the information relating to the visual identifier **16** to the controller **18**. The remote device **20** may also communicate image data relating to the surrounding environment to the controller **18**.

(84) In decision step **206**, the controller **18** receives and analyzes the information related to the visual identifier **16** from the remote device **20**. Each visual identifier **16** is unique. For example, each visual identifier **16** is associated with a location, which corresponds with a specific treatment device **14**, room environment **40**, and/or patient. The location may be the identifying or unique information (e.g., the configuration) in the visual identifier **16** that allows the information system **10** to recognize and correlate the specific information to be retrieved with the visual identifier **16**. In a non-limiting example, the visual identifier **16** on the room plaque **180** is associated with the specific patient room **40**A, the patient in the patient room **40**A, or both. In another non-limiting example, the visual identifier **16** on a patient identification bracelet (e.g., the identification feature **38**) is associated with the specific patient. In decision step **206**, the controller **18** determines whether the location of the visual identifier **16** is recognized.

(85) If the location of visual identifier **16** is not recognized in decision step **206**, the information system **10** proceeds to step **208**. In step **208**, the controller **18** generates the virtual image **30**, which includes more general, non-confidential information. This more general information may include device information **26**, such as a manufacturer, charge level, location of the treatment device **14** within the medical facility **12** (e.g., floor **42**, unit, etc.), as well as more general information about the medical facility **12**, such as facility name, location of nurse call stations, and location of exits. Without recognition of the location, the information system **10** may not associate the specific information about the treatment device **14** or the patient with the visual identifier **16**.

(86) Returning to decision step **206**, if the controller **18** recognizes the location, the information system **10** proceeds to decision step **210** to determine whether the caregiver with the remote device **20** has authorized access to the information associated with the location of the visual identifier **16**. The controller **18** communicates with remote device **20**, which includes the authorized access interface **168**. If the caregiver has been granted access via the authorized access interface **168**, touch identification feature **170**, facial recognition, or iris authentication, the controller **18** recognizes the authorization of the caregiver to access some of all of the information.

(87) There may be different levels of access for different caregivers, which may be based on the individual caregiver, the role of the caregiver, the seniority of the caregiver, etc. The level of access granted to each caregiver or each type of caregiver (e.g., nurse, technician etc.) may be stored within the memory **56** of the information system **10**. The level of access may be associated with the stored identification information (e.g., credentials, fingerprint information, stored facial or iris image, etc.). The controller **18** may then grant access to the authorized information or options, and may not include any unauthorized information or options within the virtual image **30**. Accordingly, the virtual image **30** may be different for caregivers having different authorization levels. If the

user is not authorized for any of the information associated with the specific visual identifier **16**, the information system **10** proceeds to step **208** to display the more general information.
(88) Returning to decision step **210**, if the caregiver is authorized to view or access at least some of the information associated with the visual identifier **16**, the information system **10** proceeds to step **212** of generating the virtual image **30** to include at least one of the patient information **48**, the device information **26** for treatment devices **14** in the room environment **40**, and information associated with the room environment **40** (e.g., room information). The virtual image **30** is generated and communicated to the remote device **20**. The patient information **48** and/or room information associated with the room plaque **180** is then displayed to the caregiver. The virtual image **30** with the patient information **48** and/or the room information may be overlaid on the captured image data and displayed via the handheld device **80**, may be projected into the field of view **116** of the caregiver using the wearable device **100**, or a combination thereof. Accordingly, the virtual signage having confidential information may be displayed to the caregiver, thereby being viewable by the caregiver and not others in the surrounding area. The confidential information may include information helpful for the caregiver to know prior to entry into the room environment **40**.
(89) In step **214**, the caregiver may enter the room environment **40**, such as the patient room **40**A, associated with the room plaque **180**. The caregiver may position the remote device **20** to sense at least one subsequent visual identifier **16** within the room environment **40**. The visual identifier **16** may be associated with the patient, with at least one treatment device **14**, the room environment **40** in general, or other objects or systems within the room environment **40**.
(90) In decision step **216**, the controller **18** receives the information relating to the subsequent visual identifier **16** and determines whether the location of the visual identifier **16** is recognized, similar to decision step **206**. If the location is not recognized, the information system **10** proceeds to step **208** of displaying the virtual signage with more general information. Without the location being recognized, the controller **18** may not be able to determine the information associated with the visual identifier **16** to be retrieved.
(91) If in decision step **216** the location of the visual identifier **16** is recognized, the information system **10** proceeds to step **218** of generating the virtual image **30** and displaying the virtual image **30** (e.g., the virtual signage) to the caregiver. The controller **18** may utilize the access level of the caregiver determined in decision step **210** and display the information according to the access level of the caregiver. Alternatively, the controller **18** may confirm the access level of the caregiver in step **218**. The access level may differ for each visual identifier **16**.
(92) Additionally, in step **218**, the caregiver may view or interact with the virtual image **30** to obtain the desired information about the patient, the treatment device **14**, the room environment **40**, etc. The virtual image **30** may also include facility protocols **126** and prompts to activate facility protocols **126** based on the information from the treatment device **14**, the electronic medical records **120**, or the caregiver interaction system **122**. Additionally, the virtual image **30** may provide a prompt for action based on or aligned with the facility protocols **126**.
(93) Further, in step **218**, the controller **18** may generate additional virtual images **30** in response to recognizing other visual identifiers **16** within the surrounding environment. Additionally or alternatively, the controller **18** may generate subsequent or updated virtual images **30** based on the interaction of the caregiver with the virtual image **30**, new information available and retrieved by the controller **18**, the position of the caregiver relative to the virtual image **30**, or a combination thereof. The updated virtual image **30** may replace a previous virtual image **30** or supplement the previous virtual image **30** to include additional information. The virtual image **30** may be automatically updated in response to sensed information, at predefined intervals, etc.
(94) In step **220**, the virtual image **30** may be stored within the memory **56** of the controller **18**, the remote device **20**, the electronic medical record **120** of the patient, or a combination thereof. Additionally, the virtual image **30** may be shared or communicated to other caregivers via the

caregiver interaction system **122**. When the virtual image **30** has been viewed by the caregiver, the caregiver may terminate the display of the virtual image **30**, and the method **200** ends at **222**. It is understood that the steps **202-222** of the method **200** may be performed in any order, simultaneously, repeated, and/or omitted without departing from the teachings provided herein.
(95) With reference to FIGS. **1-13**, the information system **10** provides an efficient process for the caregiver to view information from a variety of sources **28** simultaneously, while the confidentiality of the information is maintained. The caregiver views the device information **26**, the patient information **48**, and/or the room information in the virtual image **30** via augmented reality or mixed reality. The information system **10** provides selective access to the information, which heightens patient privacy. The virtual image **30** is generated in response to specific visual identifiers **16** or other data tags identified by the information system **10**. The information system **10** compiles information from the treatment devices **14**, the electronic medical record **120**, the caregiver interaction system **122**, the local server **50**, the remote server **52**, the facility protocols **126**, the sensor assembly **140**, and any other system or device associated with the medical facility **12**. In this way, the virtual image **30** provides a single location for the caregiver to view information that is useful for treating and caring for the patient. The information system **10** also provides multiple virtual images **30** to provide the information to the caregiver in updated, supplemented, or new virtual images **30**.
(96) The data tag configured as the visual identifier **16** may be any optically recognizable feature identifiable by the information system **10**. It is contemplated that other configurations of the data tag using additional or alternative modes of sensing information, such as radio frequency identification (RFID) tags, location tags, etc., may be utilized by the information system **10** to determine the information to be provided to the caregiver without departing from the teachings herein. The information system **10** may include one or several modes of sensing information.
(97) Use of the present system may provide for a variety of advantages. For example, the caregiver may retrieve information from a variety of locations or sources **28** using the information system **10**, which increases the efficiency of the caregiver in treating the patient. Additionally, the caregiver may view the information in at least one virtual image **30** generated by the information system **10** and displayed directly to the caregiver. Further, the caregiver may provide increased contactless care by utilizing the information system **10**, rather than using hard copies of handwritten notes and other hardware. Also, the information system **10** may utilize augmented reality to overlay the virtual image **30** on image data to be displayed by the handheld device **80**. Additionally, information system **10** may use mixed reality to project the virtual image **30** into the field of view **116** of the caregiver wearing the wearable device **100**. Further, when utilizing the wearable device **100**, the caregiver may interact with or manipulate the virtual image **30**. Moreover, the information system **10** may provide greater privacy to the patient. The information system **10** may also reduce human-based errors, by populating information into the virtual image **30** automatically. Additionally, the information system **10** may increase convenience and accessibility to data regardless of where the data is stored. Additional benefits or advantages of using this system may be realized and/or achieved.
(98) The memories disclosed herein may be implemented in a variety of volatile and nonvolatile memory formats. The controller and control units described herein may include various types of control circuitry, digital or analog, and may include a processor, a microcontroller, an application specific integrated circuit (ASIC), or other circuitry configured to perform various inputs or outputs, control, analysis, and other functions described herein.
(99) The device and method disclosed herein is further summarized in the following paragraphs and is further characterized by combinations of any and all of the various aspects described therein.
(100) According to at least one aspect of the present disclosure, an information system for displaying virtual signage in a medical facility includes a treatment device. A visual identifier is operably coupled to the treatment device and a controller is configured to communicate with a

remote device having a sensor for sensing the visual identifier within a field of detection. The controller is configured to recognize the visual identifier sensed by the remote device, determine device information associated with the visual identifier based on a configuration of the visual identifier, retrieve the device information relating to the treatment device associated with the visual identifier from an information source, and generate a virtual image including the device information configured to be viewed via the remote device.

(101) According to another aspect, a treatment device is at least one of a mattress, a medical bed, a surgical table, and a vital signs monitor.

(102) According to another aspect, a controller is configured to retrieve patient information from an electronic medical record stored in a server. A virtual image includes the patient information from the electronic medical record.

(103) According to another aspect, a controller is configured to overlay a virtual image on image data sensed by a remote device. The controller is configured to communicate the virtual image combined with the image data to the remote device to be displayed.

(104) According to another aspect, a controller is configured to communicate a virtual image to a remote device to be projected into a field of view of a user wearing the remote device.

(105) According to another aspect, a controller is configured to communicate with a sensor of a remote device that is configured to sense at least one of movement of a user of the remote device and a focus direction of the user. The controller is configured to generate an updated virtual image in response to at least one of the movement and the focus direction of the user.

(106) According to another aspect, a controller is configured to determine an access level of a user via identification information received via at least one of an authorized access interface on a remote device, a touch identification feature on a remote device, and a sensor on the remote device to determine if the user has authorized access to device information.

(107) According to another aspect, a remote device is at least one of a handheld device having a display and a wearable device having a projector. The remote device is configured to utilize at least one of augmented reality and mixed reality to display a virtual image to a user.

(108) According to another aspect, an information system for a medical facility in which visual identifiers are associated with the medical facility where the information system includes a query member associated with each visual identifier. The query member is at least one of a patient identification feature, a treatment device, and a room environment. A controller is configured to communicate with a remote device and a server. The controller is configured to recognize the visual identifiers sensed by the remote device, associate the visual identifiers with information related to at least one of a patient, the treatment device, and the room environment based on the query member, retrieve the information from an information source, and generate a virtual image including the information to be communicated to a user via the remote device.

(109) According to another aspect, an information source is at least one of a local server, a remote server, a sensor assembly of a treatment device, and a caregiver interaction system.

(110) According to another aspect, each visual identifier is coupled to at least one of a room plaque associated with a room environment, a patient identification feature associated with a patient, a treatment device, and a surface of the room environment.

(111) According to another aspect, a controller is configured to retrieve information related to a patient from an electronic medical record stored in a server.

(112) According to another aspect, a controller is configured to compare information with a facility protocol stored in a server. A virtual image includes at least one of the facility protocol and a prompt for action aligned with the facility protocol.

(113) According to another aspect, a remote device is a handheld device having a display and an imager configured to capture image data within a field of detection. A controller is configured to overlay a virtual image over the image data captured by the imager to communicate the information to a user via the display.

(114) According to another aspect, a remote device is a wearable device that includes a projector configured to project a virtual image to communicate information to a user.
(115) According to another aspect, a wearable device includes environmental sensors configured to sense environmental information in a surrounding area and user sensors configured to sense at least one of movement and focus direction of a user. A controller is configured to update a virtual image to a subsequent virtual image in response to at least one of movement of the user and focus direction of the user.
(116) According to another aspect, a controller is configured to determine an access level of a user based on identification information received by a remote device. Information in a virtual image is based on the access level.
(117) According to another aspect, a method of displaying information to a caregiver includes sensing a visual identifier positioned within a field of detection of a sensor of a remote device via at least one of an imager and an environmental sensor; recognizing the visual identifier; retrieving information relating to at least one of a room environment, a patient, and a treatment device based on the configuration of the visual identifier; generating a virtual image including the information; and displaying the virtual image via at least one of a display of the remote device and within a field of view of a user of the remote device.
(118) According to another aspect, a method includes coupling a visual identifier to at least one of a room environment, a treatment device, a patient identification feature, and a room plaque and positioning a remote device such that the visual identifier is within a field of detection.
(119) According to another aspect, a method includes determining an access level of a user. A step of generating a virtual image includes generating the virtual image to include information according to an access level of the user.
(120) A means for conveying information to a caregiver at a medical facility includes a means for identifying associated with a room environment of the medical facility. A means for controlling is configured to communicate with a remote device and a server. The means for controlling is configured to recognize the means of identifying sensed by the remote device, associate the means for identifying with patient information stored in the server. And generate a virtual image including the patient information to be communicated to the caregiver via a remote device.
(121) Related applications, for example those listed herein, are fully incorporated by reference. Descriptions within the related applications are intended to contribute to the description of the information disclosed herein as may be relied upon by a person of ordinary skill in the art. Any changes between any of the related applications and the present disclosure are not intended to limit the description of the information disclosed herein, including the claims. Accordingly, the present application includes the description of the information disclosed herein as well as the description of the information in any or all of the related applications.
(122) It will be understood by one having ordinary skill in the art that construction of the described disclosure and other components is not limited to any specific material. Other exemplary embodiments of the disclosure disclosed herein may be formed from a wide variety of materials, unless described otherwise herein.
(123) For purposes of this disclosure, the term “coupled” (in all of its forms, couple, coupling, coupled, etc.) generally means the joining of two components (electrical or mechanical) directly or indirectly to one another. Such joining may be stationary in nature or movable in nature. Such joining may be achieved with the two components (electrical or mechanical) and any additional intermediate members being integrally formed as a single unitary body with one another or with the two components. Such joining may be permanent in nature or may be removable or releasable in nature unless otherwise stated.
(124) It is also important to note that the construction and arrangement of the elements of the disclosure, as shown in the exemplary embodiments, is illustrative only. Although only a few embodiments of the present innovations have been described in detail in this disclosure, those

skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes, and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter recited. For example, elements shown as integrally formed may be constructed of multiple parts, or elements shown as multiple parts may be integrally formed, the operation of the interfaces may be reversed or otherwise varied, the length or width of the structures and/or members or connector or other elements of the system may be varied, the nature or number of adjustment positions provided between the elements may be varied. It should be noted that the elements and/or assemblies of the system may be constructed from any of a wide variety of materials that provide sufficient strength or durability, in any of a wide variety of colors, textures, and combinations. Accordingly, all such modifications are intended to be included within the scope of the present innovations. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions, and arrangement of the desired and other exemplary embodiments without departing from the spirit of the present innovations.
(125) It will be understood that any described processes or steps within described processes may be combined with other disclosed processes or steps to form structures within the scope of the present disclosure. The exemplary structures and processes disclosed herein are for illustrative purposes and are not to be construed as limiting.

Claims

1. An information system for displaying virtual signage in a medical facility, comprising: a treatment device including a sensor assembly configured to sense device information about the treatment device and patient information about a patient using the treatment device; a visual identifier operably coupled to the treatment device; and a controller configured to communicate with a remote device having a sensor for sensing the visual identifier within a field of detection, wherein the controller is configured to: recognize the visual identifier sensed by the remote device; determine device information associated with the visual identifier based on a configuration of the visual identifier; receive sensed device information relating to the treatment device and sensed patient information relating to the patient associated with the visual identifier from the treatment device; compare at least one of the sensed device and patient information with at least one of a fall risk protocol and a pulmonary risk protocol; generate a virtual image including the sensed device and patient information and risk information related to the at least one of the fall risk protocol and the pulmonary risk protocol configured to be viewed via the remote device; determine a direction of focus for eyes of a user; determine a field of view of the user based on the direction of focus for the eyes; project the virtual image as a projected virtual image via a projector into a surrounding environment within the field of view of the user; determine a user interaction with the projected virtual image; determine a selection based on the user interaction with the projected virtual image; project an updated projected virtual image into the surrounding environment via the projector based on a sensed user interaction with the projected virtual image; and adjust the treatment device in response to the selection made via the projected virtual image.
2. The information system of claim 1, wherein the treatment device is at least one of a mattress, a medical bed, a surgical table, and a vital signs monitor.
3. The information system of claim 1, wherein the controller is configured to retrieve patient information from an electronic medical record stored in a server, and wherein the virtual image includes the patient information from the electronic medical record.
4. The information system of claim 1, wherein the controller is configured to communicate with the sensor of the remote device that is configured to sense at least one of movement of the user of the remote device and the direction of focus for the eyes of the user, wherein the controller is

configured to generate the updated projected virtual image in response to at least one of the movement and a change in the direction of focus for the eyes of the user.

5. The information system of claim 1, wherein the controller is configured to determine an access level of the user via identification information received via at least one of an authorized access interface on the remote device, a touch identification feature on the remote device, and the sensor on the remote device to determine if the user has authorized access to the device information.

6. The information system of claim 1, wherein the remote device is a wearable device having the projector, and wherein the remote device is configured to utilize at least one of augmented reality and mixed reality to display the virtual image to the user.

7. The information system of claim 1, wherein the remote device is a wearable remote device that includes environmental sensors configured to sense environmental information in a surrounding area and user sensors configured to sense movement of the user, wherein the controller is configured to update the virtual image to the updated projected virtual image in response to at least one of the movement of the user and a change in the direction of focus of the user.

8. The information system of claim 1, wherein the controller is configured to: receive at least one of call information from a nurse call system and interaction information from a caregiver interaction system; and generate the virtual image including at least one of the call information from the nurse call system and the interaction information from the caregiver interaction system.

9. An information system for a medical facility in which visual identifiers are associated with said medical facility, the information system comprising: a query member associated with each visual identifier, wherein the query member is at least one of a patient identification feature, a treatment device, and a room environment; and a controller configured to communicate with a remote device and a server, wherein the controller is configured to: receive a picture from the remote device including at least one visual identifier and a surrounding environment proximate the at least one visual identifier, the picture being image data of a real-world environment with at least one physical object in the surrounding environment; recognize the at least one visual identifier in the picture; associate the at least one visual identifier with information related to at least one of a patient, the treatment device, and the room environment based on the query member; retrieve the information from multiple information sources including the server and at least one of a nurse call system and a caregiver interaction system; generate a virtual image including the information, the information including at least one of sensed information, risk information relating to at least one facility protocol, call information from the nurse call system, and interaction information from the caregiver interaction system; overlay the virtual image on the image data of the picture to augment the picture and form a combined image depicting the information in the real-world environment with the at least one physical object in the surrounding environment; and communicate the combined image to a user via a display of the remote device.

10. The information system of claim 9, wherein the server is at least one of a local server and a remote server, and wherein the multiple information sources includes at least one of the local server, the remote server, a sensor assembly of the treatment device, and the caregiver interaction system.

11. The information system of claim 9, wherein each visual identifier is coupled to at least one of a room plaque associated with the room environment, the patient identification feature associated with the patient, the treatment device, and a surface of the room environment.

12. The information system of claim 9, wherein the controller is configured to retrieve the information related to the patient from an electronic medical record stored in the server.

13. The information system of claim 9, wherein the controller is configured to compare the information with a facility protocol stored in the server, and wherein the virtual image includes at least one of the facility protocol and a prompt for action aligned with the facility protocol.

14. The information system of claim 9, wherein the remote device is a handheld device having the display and an imager configured to capture the picture within a field of detection.

15. The information system of claim 9, wherein the controller is configured to determine an access level of the user based on identification information received by the remote device, and wherein the information in the virtual image is based on the access level.
16. The information system of claim 9, wherein the display is a display screen, and wherein the controller is configured to communicate the combined image to the display screen of the remote device as a combined augmented picture.
17. A method of displaying information to a caregiver, comprising: sensing a visual identifier positioned within a field of detection of a sensor of a remote device via at least one of an imager and an environmental sensor; recognizing the visual identifier; receiving information relating to at least one of a room environment, a patient, and a treatment device based on a configuration of the visual identifier, the information including sensed data related to the treatment device and the patient; receiving at least one of call information from a nurse call system and interaction information from a caregiver interaction system; comparing the sensed data with facility protocols, the facility protocols including at least one of safety protocols, device protocols, and risk information; prompting an action based on the facility protocols in response to the sensed data; generating a virtual image including the information, the sensed data, the risk information, and at least one of the call information and the interaction information, the virtual image including the action based on the at least one of the safety protocols and the device protocols; displaying the virtual image via at least one of a display of the remote device and within a field of view of a user of the remote device; determining a user interaction with the virtual image; monitoring a user position relative to the virtual image; generating at least one of a new virtual image and an updated virtual image based on the user interaction and the user position relative to the virtual image; and displaying the at least one of the new virtual image and the updated virtual image.
18. The method of claim 17, further comprising: coupling the visual identifier to at least one of the room environment, the treatment device, a patient identification feature, and a room plaque; and positioning the remote device such that the visual identifier is within the field of detection.
19. The method of claim 17, further comprising: determining an access level of the user, wherein the step of generating the virtual image includes generating the virtual image to include the information according to the access level of the user.
20. The method of claim 17, wherein the step of displaying the virtual image includes projecting the virtual image via a projector to form a projected virtual image into a surrounding environment, and wherein the step of determining the user interaction includes determining a selection based on the user interaction with the projected virtual image, the method further comprising: adjusting the treatment device in response to the selection made via the projected virtual image.
